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Palaeomagnetic results from the southern Sierra Madre Oriental, Mexico: evidence for Early Cretaceous or Laramide remagnetization?

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ABSTRACT

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A large suite of samples from the Latest Triassic Huizachal and Early Jurassic Huayacocotla Groups and the Latest Jurassic Taman and Earliest Cretaceous Pimienta Formations was collected in the southern Sierra Madre Oriental for a palaeomagnetic study. Only the samples from three sites belonging to the Huizachal Group and the Las Juntas Formation possibly have retained their primary magnetization. If so, their pole position does not reveal any palaeomagnetically discernable motion relative to cratonic North America. All remaining sites were remagnetized as evidenced by a negative fold test at four sites and the fact that the pole positions cluster better at the 95% significance level if no structural corrections are applied. The tightness of the cluster ($\alpha_{95} = 4.6°$) and the same polarity, suggest that the samples were remagnetized at some common time. These results permit two interpretations. (1) If the sampling region has not suffered any significant tectonic rotation, then the remagnetization can be dated by comparison with the polar wander path for North America as Early Cretaceous (≈ 130 Ma). This Early Cretaceous phase of deformation is not recognized in the northern Sierra Madre Oriental and clearly pre-dates the Early Tertiary Laramide orogeny. (2) If the southern Sierra Madre Oriental did rotate counterclockwise by $\approx 20°$, then the remagnetization could have originated in any Cretaceous or Early Tertiary time, and may indeed be related to the Laramide deformation. In either case, the data imply that the southern Sierra Madre Oriental constitutes an independent tectonic domain.

1. Introduction

The Sierra Madre Oriental (SMO) is the dominant topographic feature of eastern Mexico and is generally interpreted as an Early Tertiary "Laramide" fold belt (De Cserna, 1976; Tardy, 1977, 1980; Belcher, 1979). Palaeomagnetic results from Triassic/Jurassic rocks of the northern SMO in the Ciudad Victoria area (Belcher, 1979; Gose et al., 1982) imply large counterclockwise rotations ($\approx 125°$) seemingly related to the easterly motion of Mexico along major faults such as proposed by De Cserna (1976) and Anderson and Schmidt

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(1983) among others. (see compilation in Urrutia Fucugauchi, 1984). In the Arteaga Canyon near Saltillo, Kleist (1980) sampled the Lower Cretaceous Cupido limestone. The samples were apparently remagnetized in post-Middle Aptian time but prior to the Laramide deformation. The data suggest a 35–40° counterclockwise rotation which Kleist interprets to be the result of "wrapping of the folds around the southeast corner of the ancient Coahuila Peninsula, which served as a barrier to the Laramide decollement deformation". Similar results were obtained by Bonfiglio (1982) who sampled the Albian Aurora limestone at the same locality.

This paper reports the results from two independent studies in the southern SMO. The Huizachal Formation and Huayacocotla Group were sampled and analyzed by Böhnel (1985) with the goal of obtaining data relevant to the tectonic evolution of the Gulf of Mexico Basin. Gose and Testarmata (1984) sampled the Taman and Pimienta formations for a magnetostratigraphic study. Subsequently, limestones belonging to the Pimienta and Tamaulipas Inferior Formation, which are overlaying discordantly the Huayacocotla Group, were studied for magnetostratigraphy.

2. Geology

The sediments of the Huizachal and Huayacocotla Groups are exposed in a relatively undisturbed section on the east flank of the Huayacocotla Anticlinorium near Tenango de Doria, Hidalgo (Fig. 1), where they reach a total thickness of about 1750 m. Schmidt-Effing (1980) divided the Huayacocotla Group into five stratigraphic units which are, in ascending order, the Las Juntas, Temaxcalapa, Despi, and Tenango

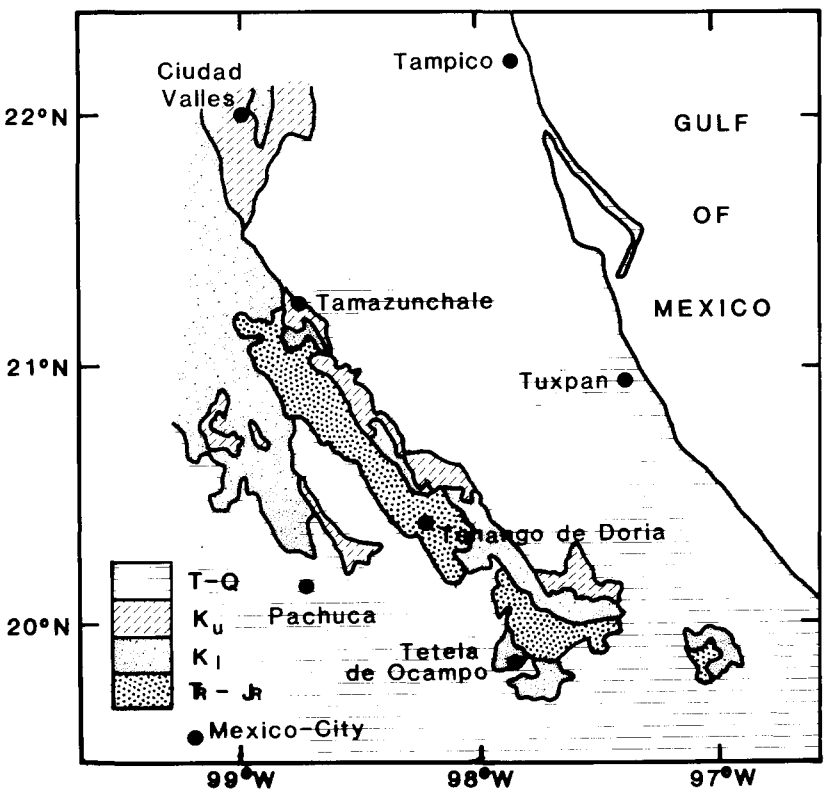


Fig. 1. Sampling area for paleomagnetic study. Simplified geology is based on Carta Geologica de la Republica Mexicana, scale 1: 2,000,000; 4th edn. (1976).

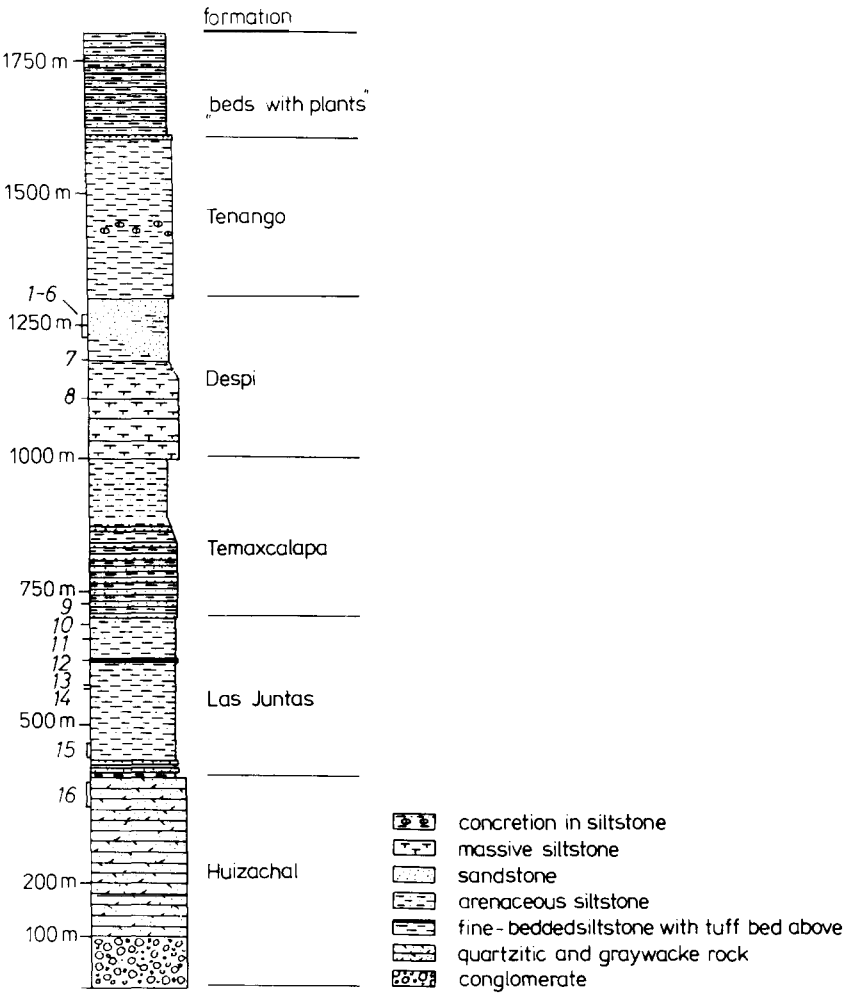


Fig. 2. Stratigraphic column of the Huizachal and Huayacocotla Groups (after Schmidt-Effing, 1980). Italic numbers indicate position of palaeomagnetic sampling sites.

Formations and the "beds with plants" (Fig. 2). Based on marine fossils, the Las Juntas through Tenango Formations are of Sinemurian age. The Huizachal Group conformably underlies the Las Juntas Formation and is thus of Hettangian and perhaps Rhaetian age at this locality. The "beds with plants" may be Lower Pliensbachian. Folding is gentle at the base of the section and increases upward, probably reflecting the competency of the strata. Paleomagnetic samples were taken at 16 sites.

These sections are discordantly overlain by limestones which are very well exposed over several hundreds of metres along deep canyons (Rio Tenango, Rio Pantepec). These limestones are sampled in 8 sites. At three of the sites, the index fossils *calpionella alpina* and *calpionellopsis oblonga* were identified in thin sections which indicate a Thithonian to Barresian–Valanginian age. The rocks belong to the Pimienta and Tamaulipas Inferior formations, which apparently are in concordant contact in the sampling region, and which are of Thithonian to Barresian–Valanginian age. Folding is generally gentle. To avoid confusion with the sites from the Pimienta formation described below, we will refer to the Pimienta and

Tamaulipas Inferior sites as Tenango limestones.

The Pimienta and Taman Formation are also well exposed along Highway 85 south of Tamazunchale in the state of Hidalgo (Fig. 1). The geology of this area has been described by Bodenlos (1956) and Longoria (1984). Much of the section is tightly folded, with chevron folds being particularly common. According to Pessagno et al. (1984), the Taman Formation ranges in age from Lower Kimmeridgian to uppermost Tithonian (Late Jurassic), and the Pimienta Formation is of Berriasian age (earliest Cretaceous). In this area the Taman Formation was sampled at four sites and the Pimienta Formation at one site. One additional outcrop of the Taman Formation was sampled about 200 km southeast of this area near the village of Tetetla de Ocampo. Here, the strata are gently dipping (8°). All limestones originated at bathyal depths. By contrast, the Huizachal and the Huayacocotla sediments are continental and shallow marine deposits.

3. Sampling and laboratory procedures

All samples were collected using standard paleomagnetic techniques. In the Huayacocotla section, samples were taken within one sedimentary bed as well as in stratigraphic order covering an entire outcrop. According to the exposure condition and rock fabric, between 6 and 63 samples were obtained. Most sites were located along river beds as here the rocks were found to be the least weathered. A special effort was made to tie in all paleomagnetic sites with the profile described by Schmidt-Effing (1980). The Tenango limestones were sampled for a preliminary magnetostratigraphic study, in a similar manner as described above. The Taman and Pimienta samples were collected in stratigraphic sequence at an average spacing of one metre. The same sections were simultaneously sampled for an analysis of their marine microfossil content (Pessagno et al., 1984).

The remanent magnetizations were measured with a spinner magnetometer as well as different cryogenic magnetometers. Rockmagnetic analyses were done with a Bison and Minisep susceptibility bridge, a pulse magnetizer constructed at the Institute of Geophysics, Münster, F.R.G. (maximum field during ~ 3 m s^{-1}, 1200 kA m^{-1}), and the high- and low-temperature isothermal remanent magnetization was studied with a Digico magnetometer. The Taman and Pimienta samples were stored in a magnetically shielded room where they remained throughout the experimental procedure. This proved to be crucial because these samples were magnetically very viscous.

4. Magnetic properties

Samples from the Huizachal and Huayacocotla Groups were subjected to a detailed rock magnetic and petrographic analysis (Böhnel, 1985). Polished sections were investigated by reflected light and scanning electron microscopy (SEM). With the given resolution of both instruments of about 1 μm, only non-magnetic oxides could be identified: pyrite, often in framboidal accumulations of probably biogenic origin, dissolved ilmenite, rutile and leukoxen. Irregular dissolved hematite was found in only one sample and no magnetite was detected. By contrast, rock magnetic analyses clearly identified (titano-) magnetite as the dominant magnetic phase. In (isothermal remanent magnetization (IRM)) experiments, most samples saturated in fields of less than 0.3×10^6 A m^{-1} (Fig. 3a) which is typical for magnetite. Continuous thermal demagnetization of saturation IRM yielded maximum unblocking temperatures between 500 and 540°C (Fig. 3d), indicative of low-titanium magnetite. Low temperature experiments showed a pronounced decrease of remanence around 150°C, which is typical for magnetite (Fig. 3c). The moderate recovery of remanence during rewarming points to a contribution of multidomain particles. Small values of saturation IRM argue for low volume content of such grains and may explain, therefore, that they were not observed microscopically.

The acquisition of IRM was also investigated for several samples from different outcrops of the Taman and Pimienta Formations. In all cases, magnetite was found to be the dominant magnetic mineral (Fig. 3b).

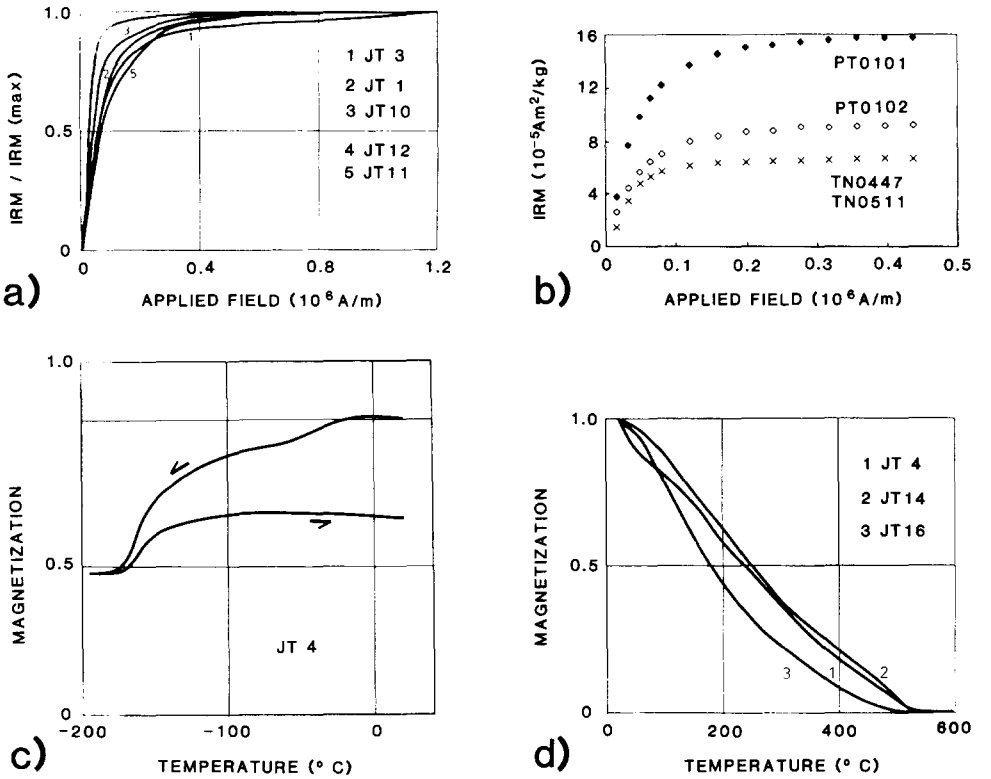


Fig. 3. Rock magnetic data of samples from sites indicated in the figure. a and b, IRM acquisition (the two TN samples gave identical results); c, low, and d, high temperature variations of room temperature strong field IRM. All experiments indicate the dominance of magnetite.

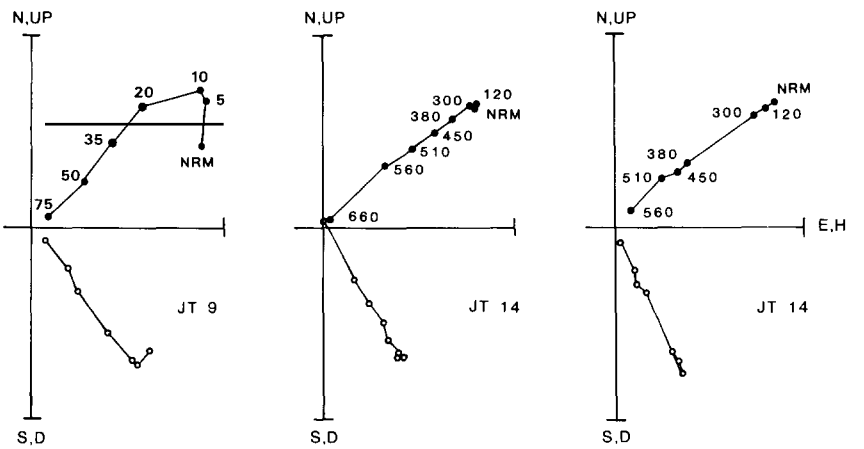


Fig. 4. Normalized orthogonal projection of magnetization vector for samples from JT sites. Open circles show the declination (NE–SW plane) and solid circles the inclination (horizontal vs. vertical component). Numbers indicate the AF demagnetization amplitude in mT or demagnetization temperature in ° C.

5. Direction of magnetization

All samples were subjected to stepwise thermal or alternating field (AF) demagnetization. In most cases, thermal demagnetization was found to be considerably more effective in isolating the characteristic remanent magnetization. The data were analyzed using modified As–Zijderveld diagrams (As, 1960; Zijderveld, 1967; Roy and Lapointe, 1978)) and the methods of Hoffman and Day (1978) or Kirschvink (1980). Samples without a stable end point could, in some cases, be successfully analyzed by the method of converging remagnetization circles (Halls, 1976). Examples of demagnetization curves are shown in Figs. 4 and 5.

5.1. Huizachal and Huayacocotla Groups (Late Triassic–Early Jurassic)

Of the 16 sites in the Huayacocotla Group, eight did not yield usable data either because the intensities of magnetization fell below the noise level of the magnetometer before defining a stable direction or because of high dispersion of directions and/or unstable magnetization directions. In the latter case, the samples often exhibited an increase in intensity as well as susceptibility near 400 °C, indicating the formation of new magnetic phases.

Table 1 lists the statistical parameters for the remaining eight sites both without and with corrections for bedding attitudes. All sites, except one horizon within site JT16, are reversely magnetized. This is rather surprising in view of the fact that the Jurassic period is characterized by very frequent reversals (see review in Ogg and Steiner, 1986) and in the sample section which represents about 10 Ma, several magnetozones should have been encountered. This suggests that most if not all of the section has been remagnetized.

Site JT16, however, may have retained its primary magnetization. A large portion of this 50-m section is of normal polarity. Admittedly, due to the low NRM intensity and rather unstable demagnetization behaviour, the normal direction is only very poorly defined and will be deleted from further considerations, but the fact that its

TABLE 1

Statistics for Huizachal and Huayacocotla groups

		N	**R**	K	α_{95} (deg)	D (deg)	I (deg)	LAT (deg)	LONG (deg)
JT04	NSC	29/5	26.90	13	7.6	142.7	−25.9	53.8	167.1
	SC	29/5	26.90	13	7.6	141.4	−44.8	54.2	189.0
JT05	NSC	18/7	16.29	10	11.5	137.8	−48.5	50.9	193.8
	SC	18/7	16.29	10	11.5	165.8	−47.2	74.8	206.2
JT09	NSC	27/7	26.69	85	3.0	150.7	−26.5	61.3	163.6
	SC	27/7	26.69	85	3.0	135.3	−33.1	47.9	176.5
JT10	NSC	11/4	9.75	81	7.2	46.0	−21.5	56.1	161.2
	SC	11/4	10.06	111	4.6	53.1	−35.6	64.8	174.9
JT12	NSC	16/0	15.65	43	5.7	153.0	−15.6	61.1	150.2
	SC	16/0	15.20	19	8.7	145.3	−29.9	56.8	170.1
JT13	NSC	19/8	16.64	81	2.9	141.6	−36.3	54.1	178.3
	SC	19/8	16.64	81	2.9	166.5	−22.6	74.4	140.2
JT14	NSC	42/5	41.57	96	2.3	155.5	−42.3	67.0	186.6
	SC	42/5	41.57	96	2.3	181.8	−18.3	78.9	72.6
JT16R	NSC	27/8	23.73	8	10.5	161.4	−13.2	67.4	137.2
	SC	27/8	23.73	8	10.5	167.5	−13.9	72.0	125.8
JT16N	NSC	21/7	16.13	4	17.9	333.8	32.5	65.1	170.0
	SC	21/7	16.13	4	17.9	350.5	34.2	80.9	163.1

NSC = not structurally corrected; SC = structurally corrected; N = number of samples included/rejected from analysis: **R** = resultant vector; K = precision parameter; α_{95} = half angle of 95% circle of confidence; D = east declination; I = inclination; LAT = north latitude; LONG = east longitude.

TABLE 2

Statistics for Tenango limestones

		N	**R**	K	α_{95} (deg)	D (deg)	I (deg)	LAT (deg)	LONG (deg)
TL1	NSC	5/1	4.96	115	7.2	150.0	−34.7	61.8	174.4
	SC	5/1	4.96	115	7.2	152.6	−46.1	64.1	192.9
TL2	NSC	6/0	5.80	26	13.5	163.1	−12.5	68.4	133.5
	SC	6/0	5.80	26	13.5	169.2	−40.0	79.7	186.9
TL3	NSC	10/1	9.81	49	7.0	139.4	−28.4	51.1	170.8
	SC	10/1	9.81	49	7.0	152.2	−11.6	59.4	147.3
TL4	NSC	7/1	6.96	173	4.6	153.3	−20.3	62.5	154.7
	SC	7/1	6.96	173	4.6	171.1	−40.2	81.3	190.0
TL5	NSC	8/0	7.67	22	12.1	158.4	−33.2	69.5	169.1
	SC	8/0	7.67	22	12.1	158.4	−46.0	69.1	195.4
TL6	NSC	7/2	6.49	12	18.2	153.5	−17.7	62.0	151.8
	SC	7/2	6.49	12	18.2	150.8	−47.5	62.2	194.8

NSC = not structurally corrected; SC = structurally corrected; N = number of samples included/rejected from analysis; **R** = resultant vector; K = precision parameter; α_{95} = half angle of 95% circle of confidence; D = east declination; I = inclination; LAT = north latitude; LONG = east longitude.

reversed direction is different from the other site directions (see Table 1) and that this massive sandstone is only gently folded suggests to us that the observed directions probably are of primary origin. This may be also valid for direction from sites 13 and 14, which are, from lithologically similar sequences, of low deformation degree. Support for this interpretation may be that sites JT 13, 14, and 16R group much better after structural correction than before. However, this is not significant at the 95% probability level and therefore we will delete these sites from further considerations.

TABLE 3

Statistical parameters for Taman and Pimienta formations

		N	**R**	K	α_{95} (deg)	D (deg)	I (deg)	LAT (deg)	LONG (deg)
TN01 ORTHO	NSC	15/0	14.96	328	2.1	147.4	−30.1	58.7	167.6
	SC	15/0	14.98	646	1.5	160.2	−45.4	71.1	192.6
TN02/03 OR	NSC	21/6	20.69	65	4.0	160.3	−34.9	71.4	168.7
	SC	21/6	19.42	13	9.3	162.7	−23.7	71.3	146.2
TN04 AF300	NSC	53/0	51.76	42	3.1	175.9	−46.5	82.5	232.2
	SC	53/0	49.56	15	5.2	145.1	−15.6	53.8	155.0
TN04 ORTHO	NSC	13/0	12.88	100	4.2	175.7	−46.6	82.3	231.6
(SUBSET)	SC	13/0	11.91	11	13.1	149.6	−19.6	58.8	155.1
TN05 ORTHO	NSC	12/3	11.93	151	3.5	156.2	−37.9	67.8	175.7
	SC	12/3	11.91	120	4.0	137.6	−1.0	43.8	150.2
TN06 ORTHO	NSC	14/1	13.83	75	4.6	151.7	−37.9	63.5	180.2
	SC	14/1	13.83	75	4.6	152.3	−45.9	63.6	193.8
PT01 ORTHO	NSC	11/6	10.77	43	7.1	144.8	−47.7	57.3	191.7
	SC	11/6	10.16	12	13.8	138.1	−37.9	51.1	179.1

NSC = not structurally corrected; SC = structurally corrected; N = number of samples included/rejected from analysis; **R** = resultant vector; K = precision parameter; α_{95} = half angle of 95% circle of confidence; D = east declination; I = inclination; LAT = north latitude; LONG = east longitude.

5.2. *Tenango limestones (Pimienta and Tamaulipas Inferior)* (Table 2)

Most samples exhibited a NRM of normal polarity, always changing to the lower hemisphere during thermal demagnetization between 200 to 310 °C. Dispersion of directions is rather high, partly due to NRM intensities near to the noise level of the spinner magnetometer. Again, the consistent polarity of remanence points to a remagnetization.

5.3. *Taman and Pimienta formations*

All samples from this suite contained a soft component of magnetization (Fig. 5) that was readily removed by mild heating (200 °C) or AF demagnetization to 5 or 10 mT. This magnetization is of recent origin as evidenced by the fact that the directions are aligned with the present geomagnetic field only if no structural correction is applied. Indeed, this component is of very recent origin because the directions are very close to the actual field direction, at the sampling sites, rather than to the axial dipole field direction and is thus a viscous remanent magnetization (Fig. 6).

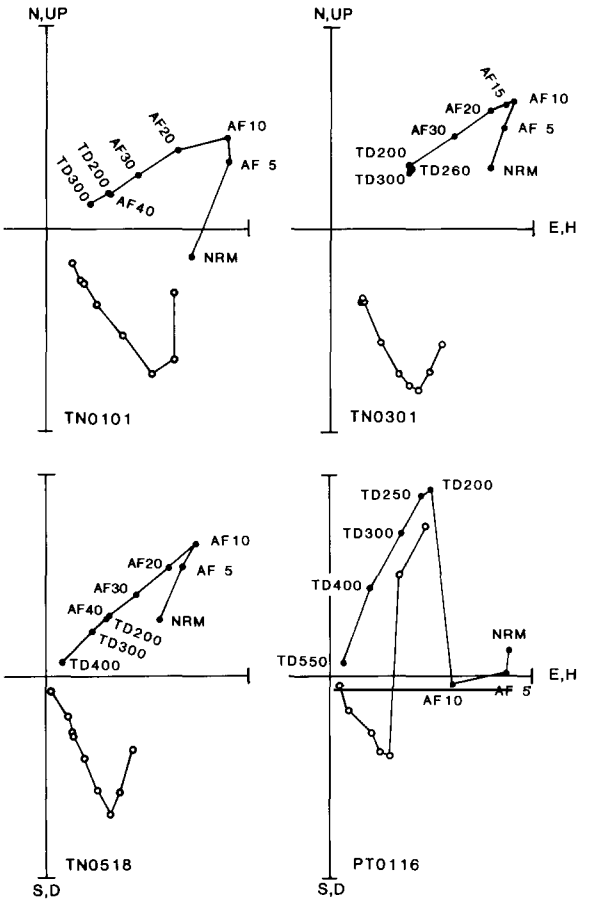


Fig. 5. Normalized orthogonal projection of magnetization vector for samples of the Taman and Pimienta Formations. Open circles represent the declination (NE–SW plane) and solid circles the inclination (horizontal vs. vertical component). The two distinct components of magnetization should be noted.

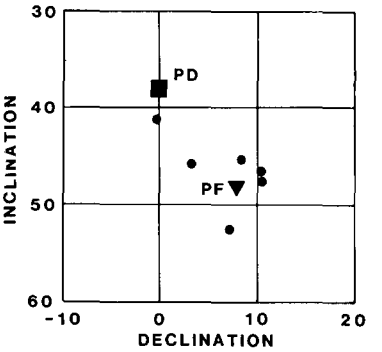


Fig. 6. Mean direction of magnetization of the low stability component for all Taman and Pimienta sites. PD = theoretical dipole field direction; PF = present field direction for sampling area. Note that the directions cluster around PF.

The response to demagnetization was consistent among most samples and statistically good data were obtained from all sites (Table 3) except one Pimienta outcrop. A fold test at three sites (TN02/3, TN04, PT01) demonstrates at the 99% confidence level that the magnetization was acquired post-folding (Fig. 7). The fact that all sites are reversely magnetized also strongly suggests remagnetization.

6. Interpretation

All sites (except JT10) where a fold test could be applied, clearly acquired their magnetization after folding; the question that needs to be addressed is whether the remaining sites are also remagnetized. Figure 8 shows the pole positions before and after structural corrections. Site TN04, while clearly remagnetized, has a pole position which is distinctively different from the other data which may indicate an unrecognized non-coaxial second phase of deformation after the original folding.

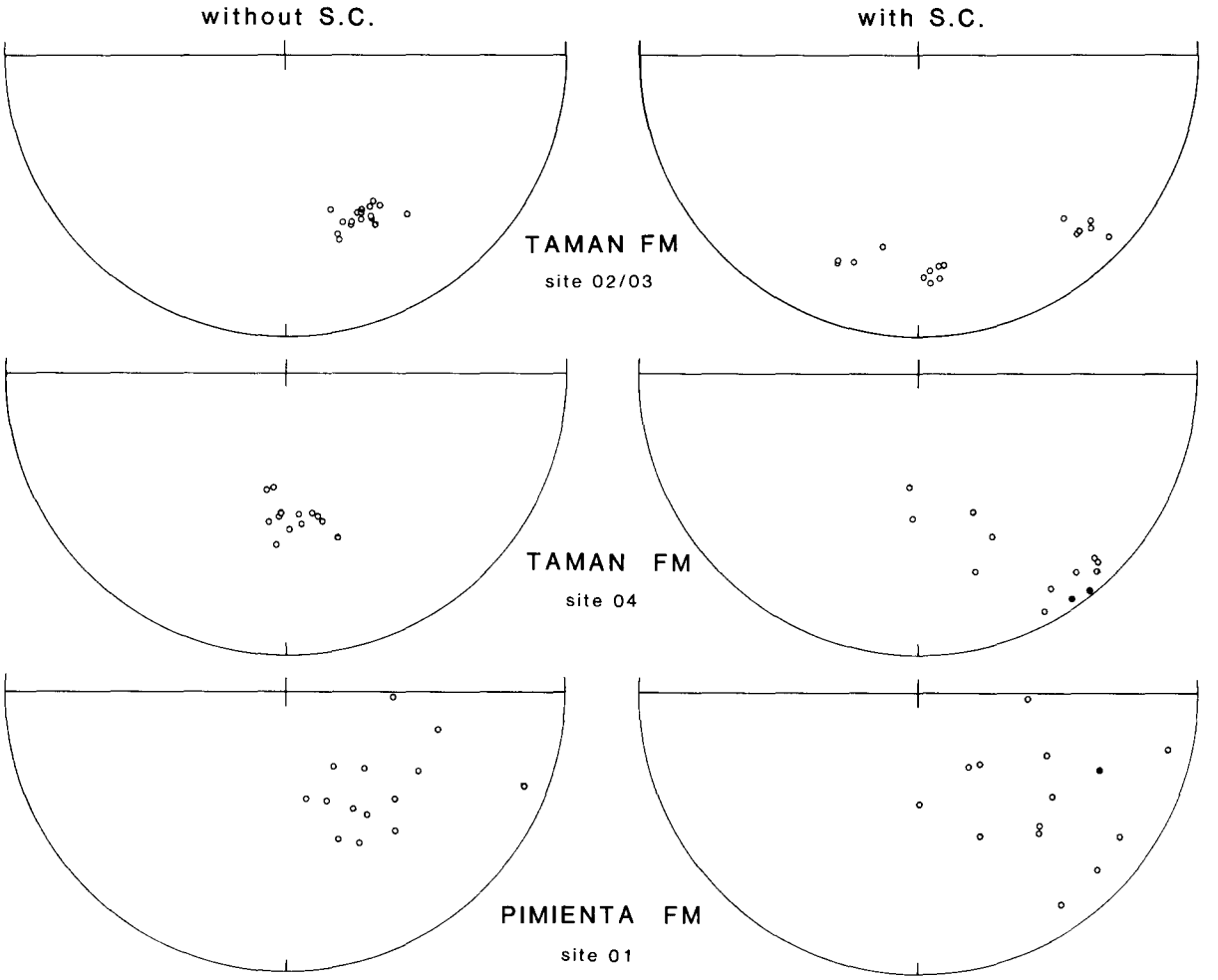


Fig. 7. Directions of stable magnetization without and with structural correction for sites TN02/03, TN04 (subset covering a 5-m fold), and PT01. Schmidt equal area projection. The better grouping without unfolding the strata implies that the magnetization post-dates the deformation.

Table 3 compares the statistical parameters of the Huayacocotla Group (without rejected sites, see above), the Tenango limestones, and the Taman and Pimienta formations (excluding site TN04) before and after tilt correction. Applying the F-test shows that for the Huayacocotla Group and for the Tenango limestones, the fold test is not significant at the 95% probability level, probably because these units have rather similar bedding attitudes. The pole positions for the other data sets cluster better at the 95% significance level if no structural correction is applied. The improvement for the combined data set is also significant at the 95% level (Table 4). The tight grouping of the uncorrected pole positions ($\alpha_{95} = 4.6°$) suggests that all rocks were remagnetized at about the same time. This is also borne out by the observation that all rocks are reversedly magnetized and no reversed epoch longer than a few million years is known to exist in Jurassic–Lower Cretaceous time (Harland et al., 1982; Ogg and Steiner, 1986). The time of remagnetization probably coincided with the final phase of deformation of these rocks, as no evidence for other thermal or orogenic events is available.

Figure 9 depicts the Cretaceous apparent polar

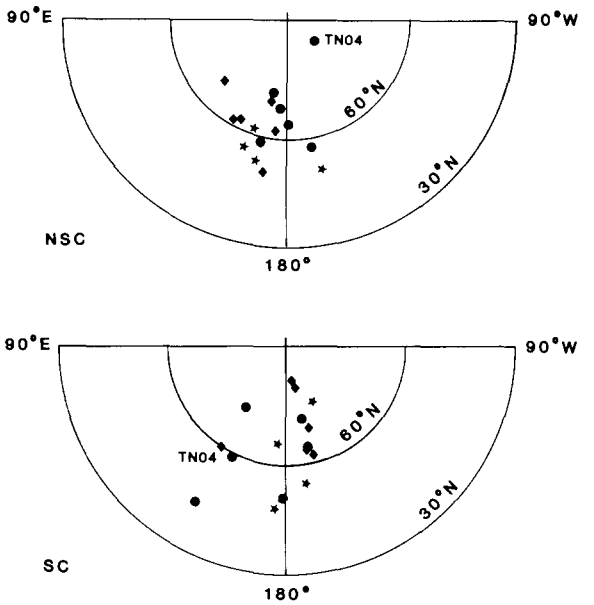


Fig. 8. Virtual geomagnetic pole positions of all sampling sites before (upper) and after (lower) applying the structural corrections. Dots, Taman-Pimienta sites; stars, Huayacocotla Group; rhombs, Tenango limestones. North polar equal area projection.

wander path (APWP) for stable North America (after Harrison and Lindh, 1982) with its 63% confidence envelope (equivalent to one standard error). Advantage of using the 63% rather than 95% confidence level is that two palaeopoles differ significantly at the 95% probability level if their standard errors do not overlap (Irving, 1979). Also shown in this figure are the pole positions of the Huizachal and Huayacocotla Groups (*), the Taman and Pimienta Formations (●), the Tenango limestones (◆), and of the combined data set (+) with their respective 63% circles of confidence. Our data coincide only with the APWP segment for the period about around 130 Ma ago, i.e. in Late Neocomian time.

7. Discussion

This study was undertaken with the aim to obtain palaeomagnetic data relevant to the tectonic evolution of Mexico and the establishment of a well-dated magnetic reversal sequence during the Late Jurassic–Early Cretaceous. With the possible exception of sites JT12 to JT16, all sites have been remagnetized and cannot be used for their intended purpose. This unexpected result is, however, of interest for the deformational history of the SMO. The event which caused the remagnetization, is clearly not a local effect as the sampling sites are distributed over some 200 km. Importantly, all sites record the same event. Because the rocks acquired their magnetization during the final phase of folding, probably the folding can be dated as post-Valanginian (post-Tamaulipas Inferior time).

The age of remanence acquisition depends on assumptions concerning the tectonic stability of north-central Mexico. If the sampling region did not experience any significant rotations since

TABLE 4

Statistics for grouped poles

		N	**R**	K	α_{95}	LAT	LONG	K_b/K_a	$F_{0.05}$
HG	NSC	4	3.96	68.6	11.2	56.3	172.2	1.83	4.28
	SC	4	3.92	37.5	15.2	60.8	183.9		
TL	NSC	6	5.93	74.5	7.8	63.2	160.4	1.59	2.98
	SC	6	5.89	46.8	9.9	70.4	182.4		
TN + PT	NSC	5	4.97	116.8	7.1	64.0	177.6	4.67	3.44
	SC	5	4.84	25.0	15.6	61.6	170.7		
HG + TL +	NSC	15	14.80	71.2	4.6	61.8	169.4	2.09	1.89
TN + PT	SC	15	14.59	34.5	6.6	65.1	178.5		

NSC = not structurally corrected; SC = structurally corrected; N = number of sites included in calculations; **R** = resultant vector; K = precision parameter; α_{95} = half angle of 95% circle of confidence; LAT = north latitude; LONG = east longitude; K_b/K_a = ratio of precision parameters before/after structural correction; $F_{0.05}$ = F-ratio for 95% probability level.

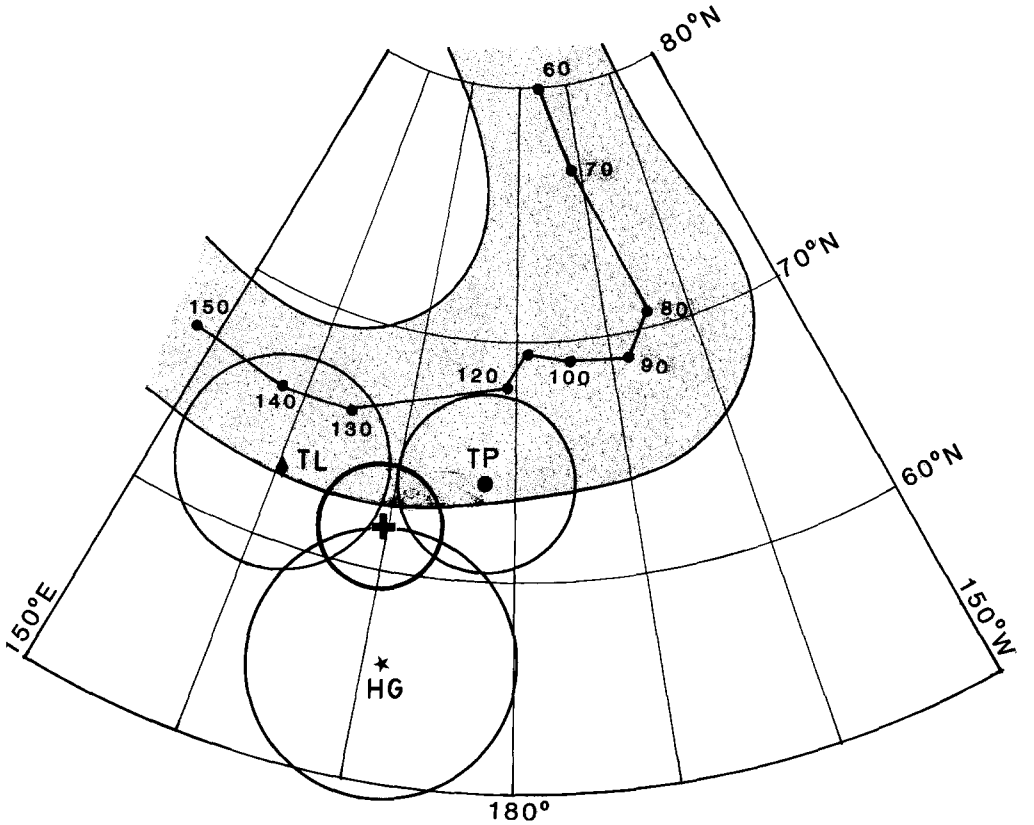


Fig. 9. Comparison of the mean pole positions for the Huizachal and Huayacocotla sites (HG). Taman and Pimienta sites (TP), and the combined data set (+) with the North American apparent polar wander path (after Harrison and Lindh, 1982) from 60 to 150 Ma. The circles and the error band represent the 63% confidence level.

Lower Cretaceous, then the position of the paleopole relative to the cratonic North American APWP implies remagnetization about 130 Ma ago (Neocomian). This is at variance with the common interpretation that the SMO was formed during the Late Cretaceous–Early Tertiary Laramide orogeny (e.g. De Cserna, 1976; Tardy, 1977, 1980). A pre-Laramide folding is supported by the observation of Bodenlos (1956) of a hiatus between the Neocomian Chapulhuacan Formation which conformably overlies the Pimienta Formation, and the Aptian–Albian Ahuacatlan Formation. Based on microfacies analysis and physical stratigraphy, Longoria (1982, 1984) confirmed the hiatus and concluded that the deformation of the southern SMO is the result of a Valanginian–Barremian tectonic episode. In addition to this Early Cretaceous event, the southern SMO also experienced a later phase of deformation as evidenced, for example, by a thrust fault just south of Tamazunchale (within our sampling area) which places the Taman, Pimienta, and Chapulhuacan Formation on top of the Late Cretaceous Xilitla Formation (Bodenlos, 1956).

The second interpretation assumes that the southern SMO did rotate in post-Tamaulipas Inferior time. At this point we do not have data from the southern SMO to support or refute this assumption. Urrutia Fucugauchi (1981) pointed out that lower Tertiary data may be interpreted in terms of a counterclockwise rotation of northern Mexico by about 20° but it is not clear whether such rotations reflect local or regional rotations (Urrutia Fucugauchi, 1984; Urrutia Fucugauchi and Böhnel, 1987). Apparent local block rotations are frequently observed along the Transmexican

Volcanic Belt which borders our sampling area to the south, and are interpreted to be caused by left lateral shear (e.g. Urrutia Fucugauchi and Böhnel, 1987, 1988). In the northern SMO, Gose et al. (1982) observed large counterclockwise rotations ($\approx 125°$) which occurred in pre-Cretaceous time. These rotations were interpreted to be the result of left-lateral shear. It thus may be reasonable to postulate that the southern SMO also rotated. Allowing for a 20° counterclockwise rotation would place the palaeopole on the Cretaceous–Lower Tertiary segment of the APWP. In this case, the remagnetization would correspond in age to the Laramide orogeny in good agreement with other interpretations such as advocated by de Cserna (1976), Tardy (1977, 1980), and others.

The existence of several major sinistral faults cutting across Mexico in a WNW–ESE direction has been postulated by several authors (see e.g. in Urrutia Fucugauchi, 1984). While the reality of these faults has never been demonstrated, palaeomagnetic data from previous studies (e.g. Gose, 1985; Urrutia Fucugauchi and Böhnel, 1987, 1988) as well as this study conceptually support this segmentation of Mexico. The data of this paper, assuming that the southern SMO did rotate, imply that this region represents a distinct tectonic domain. We note that Permian data from Chiapas in southernmost Mexico (Gose and Sanchez-Barreda, 1981) suggest that this area, too, was an independent tectonic block.

Most reconstructions of Pangaea place South America close to North America at the beginning of the Mesozoic (e.g. Dickinson, 1981; Pindell and Dewey, 1982; Anderson and Schmidt, 1983; Klitgord et al., 1984; Pindell, 1985) and require a more westerly position of Mexico for this time. Much attention has been given to the Sonora–Mojave megashear of northern Mexico, but it seems that additional faults such as envisioned by De Cserna (1976), may have played an equally important role in the tectonic evolution of Mexico.

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